

Room 6 (P6 beta2 irrationality): moderator summary

2026-09-26

Two seats (Ramanujan, Erdős), each trying a route independent of the ρ -type criteria already exhausted in `P6_note.tex`/`P6_v2_note.tex`.

Ramanujan: the regular continued fraction of q^* itself looks generic. All 999 partial quotients were computed and certified against both endpoints of the 2090-digit bracket (they agree throughout, so the sequence is correct regardless of where q^* sits inside the bracket). The maximum quotient (8250 at index 170) is a normal heavy-tailed Gauss–Kuzmin fluctuation at this length, not a Liouville-type jump; the digit-frequency histogram matches the Gauss–Kuzmin law and the geometric mean (2.7366) is close to Khinchin’s constant (2.68545); an exhaustive length-5 window scan found no periodicity or near-periodicity (which would have proved quadratic irrationality via Lagrange’s theorem, an exciting but a priori extremely unlikely outcome). **Verdict: genuine, but uninformative** – it closes off the one elementary Diophantine-approximation avenue not already covered, without being able to prove or disprove irrationality itself (a generic-looking CF is consistent with irrational, transcendental, or even a “boring” algebraic number of no special approximation quality).

Erdős: wider PSLQ search against Gamma values, q -Pochhammer/eta-type constants, Bessel zeros, and algebraic self-combinations of β_2 – nothing found. All searches returned no relation, except one which the seat itself correctly identified as spurious (a tautology from including $\beta_2 - 1$ in its own search basis) and explicitly flagged as such rather than reporting a false discovery.

Room verdict: NO PROGRESS, honestly, with two more elementary approaches closed off for precise reasons (generic CF; no exotic-constant relation found) rather than left untried. P6 stands exactly where `P6_note.tex`/`P6_v2_note.tex` left it (v10.21.0): β_2 is proved not to be a root of any integer polynomial of degree ≤ 64 ; irrationality and transcendence remain OPEN.